

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2023-24

Paper No. BSC-13A: Statistical Inference, Sampling and Design of Experiments

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) State the additive and multiplicative theorems of probability, and briefly explain their significance. [7]
- (b) Explain and define the terms: "Random Experiment", "Sample space", "Event", "Probability", "Probability Distribution", and "Statistical Hypothesis". [7]

Question 2.

- (a) Assume the random variable Y has a continuous uniform distribution defined on the interval $[0, 2]$, i.e.,

$$f(y) = \begin{cases} \frac{1}{2}, & 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Find $P(Y < 1)$.
- (ii) Find the mean of this distribution.

[7]

- (b) Define the terms "population", "sampling unit", and "sampling frame", and explain their significance in sampling theory. [7]

Question 3.

- (a) Suppose that Y is a normally distributed random variable with mean $\mu_Y = 10$ and standard deviation $\sigma_Y = 2$, and X is an independent normally distributed random variable with mean $\mu_X = 5$ and standard deviation $\sigma_X = 1$. Find the values of:
- (i) $P(Y > 12 \text{ and } X > 4)$
- (ii) $P(Y > 12 \text{ or } X > 4)$
- (iii) $P(Y > 10 \text{ and } X < 5)$

[7]

- (b) What is simple random sampling (SRS)? Explain how a simple random sample is selected from a population. What are its advantages and limitations? [7]

Question 4.

- (a) Two boxes labelled B_1 and B_2 containing candies are placed on a table. Box B_1 contains 7 red candies and 4 green candies. Box B_2 contains 3 red candies and 10 orange candies. The boxes are arranged so that the probability of selecting box B_1 is $\frac{1}{3}$ and the probability of selecting box B_2 is $\frac{2}{3}$. A blindfolded person draws a candy. What is the probability that the candy came from box B_1 given that it is a red candy? [7]
- (b) Explain the Chi-square (χ^2) test of goodness of fit. What are the key assumptions for conducting a Chi-square test? [7]

Question 5.

- (a) What is hypothesis testing, and why is it an important statistical tool in research? Discuss the concepts of null hypothesis, alternative hypothesis, Type I error, and Type II error. [6]
- (b) Weight losses (in kg) of 12 persons in an experimental one-week diet program are given as:

3.0, 1.4, 0.2, -1.2, 5.3, 1.7, 3.7, 5.9, 0.2, 3.6, 3.7, -2.0

Do these results indicate that a significant mean weight loss was achieved at the $\alpha = 0.05$ level of significance? [8]

Critical values of Student's t -distribution for one-tailed test (probability $p = 0.90$):

Degrees of Freedom (ν)	9	10	11	12	13
Critical Value ($t_{0.90,\nu}$)	1.383	1.372	1.363	1.356	1.350

Question 6.

- (a) Define 'random variable' and explain its significance in probability theory. Differentiate between discrete and continuous random variables, providing examples of each. [7]
- (b) Explain the normal distribution along with its important applications. Discuss its key properties. [7]

Question 7.

- (a) What do you understand by the binomial distribution? Discuss its key properties, including the formula for calculating binomial probabilities. [7]

- (b) What are the key steps involved in conducting stratified random sampling, and what factors should be considered when determining the strata? [7]
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Question 8. Indicate whether the following statements are **True** or **False**. If false, specify what change will make the statement true:

- (a) One of the assumptions underlying the use of the (pooled) two-sample t -test is that the samples are drawn from populations having equal means.
- (b) In the two-sample t -test, the number of degrees of freedom for the test statistic decreases as sample sizes increase.
- (c) If every observation is multiplied by 2, then the t -statistic is multiplied by 2.
- (d) The χ^2 distribution is used for making inferences about two population variances.
- (e) It is not necessary to have equal sample sizes for the paired t -test.
- (f) If the test statistic falls in the rejection region, the null hypothesis has been proven to be true.
- (g) When the test statistic is t and the number of degrees of freedom is very large ($\nu > 30$), the critical value of t is very close to that of the standard normal variable Z .

[14]